

Phishing Email Investigation Report

1. Executive Summary

This report presents a SOC Level 1 phishing email investigation. The objective was to analyze a suspicious email, identify malicious indicators, and determine whether the activity represents a phishing attempt. The investigation confirmed that the email is malicious and intended for credential harvesting through social engineering techniques.

2. Email Details

- **From:** support-paypal@security-check.com
- **To:** employee@company.com
- **Subject:** Urgent! Verify Your PayPal Account
- **Type:** Suspicious External Email

The email claims that the user's PayPal account has been limited and requests immediate verification via a provided link.

3. Email Content Analysis

The email contains multiple phishing indicators:

- Fake PayPal impersonation
- Urgency-based language ("Immediate action required")
- Suspicious external domain
- Malicious-looking URL
- Social engineering tactics to pressure the user

4. Email Header Analysis

Initial header inspection revealed:

- **SPF Result:** Fail
- **DKIM Result:** Fail
- **Authentication Status:** Failed
- **Return Path:** Spoofed sender identity detected
- **Source IP:** 185.220.101.1 (suspicious)

These failures indicate that the email was not sent from an authorized PayPal server.

5. URL Analysis

The embedded URL was analyzed using threat intelligence tools.

- **URL:** <http://paypal-login-security-check.com>

- **Result:** Malicious / Suspicious
- **Reputation:** Low trust score
- **Behavior:** Known phishing pattern (credential harvesting)

The domain impersonates PayPal and is designed to trick users into entering sensitive information.

6. Tools Used

- MXToolbox (SPF/DKIM validation)
- VirusTotal (URL reputation analysis)
- Manual email header inspection
- Basic IOC extraction techniques

7. Indicators of Compromise (IOCs)

Type	Value
IP Address	185.220.101.1
Domain	paypal-login-security-check.com
Sender Email	spoof@paypal.com
URL	http://paypal-login-security-check.com

8. Findings

- Email impersonates a trusted organization (PayPal)
- SPF and DKIM authentication failed
- Malicious domain used for phishing attempt
- Social engineering tactics detected (urgency and fear)
- High likelihood of credential harvesting attempt

9. Mitigation & Recommendations

- Block malicious domain at email gateway and firewall
- Enforce SPF, DKIM, and DMARC policies
- Enable Multi-Factor Authentication (MFA)
- Conduct phishing awareness training for users
- Monitor and filter suspicious emails using SIEM tools
- Use sandboxing for suspicious attachments and links

10. Conclusion

The email has been confirmed as a phishing attempt aimed at stealing user credentials. Proper SOC investigation techniques helped identify spoofed sender details, authentication failures, and malicious URLs. Immediate blocking and awareness measures are recommended to prevent compromise.

Analyst Note

Prepared as part of a SOC Level 1 phishing investigation exercise to demonstrate practical skills in email security analysis, threat detection, and incident reporting.